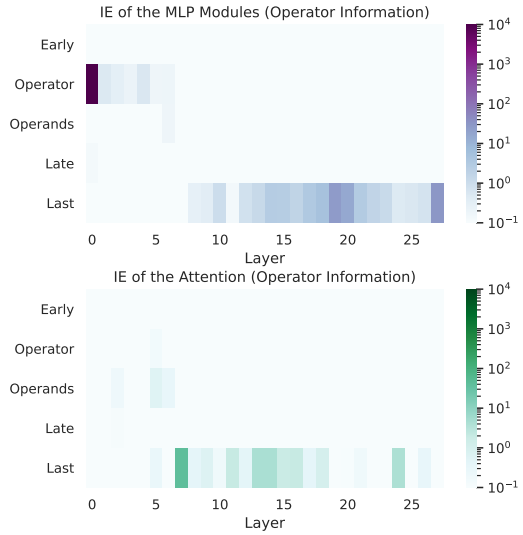

Figure 8: Indirect effect (IE) measured in GPT-J when varying the word describing the operator involved in the input query. Similar to the operands case, we observe a high contribution produced by middle-to-late MLP modules at the end of the input sequence.

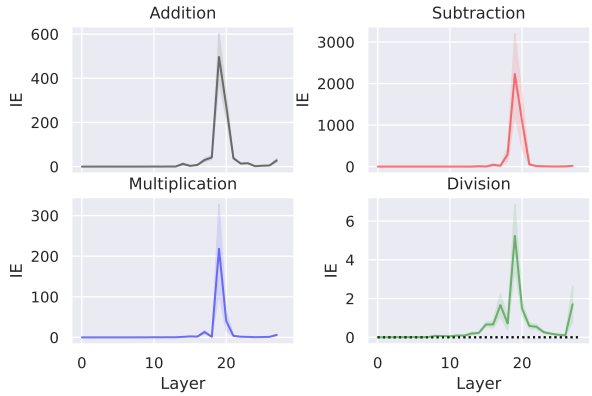

Figure 9: Indirect effect of the MLPs at the last token in each layer in GPT-J, for each of the four arithmetic operators. We observe a peak in the effect at layer 19 for all four types of operation.

A Prompt Templates

In Tables 2 and 3, we report the question templates from Karpas et al. (2022), which we used as prompts for the model for two- and three-operand queries, respectively. For three-operand queries, we use one query template for each of the 29 possible two-operation combinations.

B Performance of the Models

In Table 4, we report the accuracy of the models on the arithmetic queries that we use for our analyses. The higher accuracy obtained using numeral words is likely given by the smaller set of possible solutions considered (we used $\mathcal{S} = \{$*"one"*, *"two"*, ..., *"twenty"*$\}$, as the numeral words corresponding to larger numbers get split into multiple tokens by the tokenizer). The accuracy of GPT-J on the factual queries from the LAMA benchmark is 65.0% (we constrain the vocabulary to the set of all possible objects for all the relations considered). On the synthetic number retrieval task, GPT-J's accuracy is 86.7%.

C Flow of Operator-related Information

The measurements of the indirect effect for each model component when fixing the operand and varying the operator in the two input prompts p_1 and p_2 reveal how the model processes the information related to the operator. We report in Figure 8 the heatmap visualizations of these results for two-operand queries. Similar to the operand-related information, we observe a high effect in three activation locations: early MLP blocks corresponding to the operand tokens; middle-to-early attention modules at the last token; and middle-to-late MLP modules at the last token. These results align with our hypothesis that arithmetic-related information is transferred to the end of the sequence by the attention mechanism, where it is then processed by late MLP layers. In this setting, we measure $\mathrm{RI}(\mathcal{M}_{-1}^{\textbf{late}})$ =31.4%.

D Effects for Each Operator

For each of the four operators, we report the indirect effect measured for the last-token MLP modules in GPT-J in Figure 9. The results for each of the four operators show a common spike in the effect at layers 19-20. This indicates the presence of a specific part of the model relevant to the numerical predictions of the bi-variate arithmetic questions, irrespective of the operator involved. We also notice a difference in the magnitude of the effects, which is linked to the capability of the model to correctly answer the query.

E Changes in the Model's Prediction

We measured the influence of the model components in terms of probability changes. Now, we study the dynamics of the actual model predictions. In particular, considering the scenario in which $r = r'$, we verify whether the intervention leads to a change in the model's prediction. That is, we

Type	addition	subtraction
1	Q: How much is n_1 plus n_2? A:	Q: How much is n_1 minus n_2? A:
2	Q: What is n_1 plus n_2? A:	Q: What is n_1 minus n_2? A:
3	Q: What is the result of n_1 plus n_2? A:	Q: What is the result of n_1 minus n_2?
3	Q: What is the sum of n_1 and n_2? A:	Q: What is the difference between A: n_1 and n_2? A:
5	The sum of n_1 and n_2 is	The difference between n_1 and n_2 is
6	n_1 + n_2 =	n_1 - n_2 =
	multiplication	**division**
1	Q: How much is n_1 times n_2? A:	Q: How much is n_1 over n_2? A:
2	Q: What is n_1 times n_2? A:	Q: What is n_1 over n_2? A:
3	Q: What is the result of n_1 times n_2? A:	Q: What is the result of n_1 over n_2? A:
4	Q: What is the product of n_1 and n_2? A:	Q: What is the ratio between n_1 and n_2? A:
5	The product of n_1 and n_2 is	The ratio of n_1 and n_2 is
6	n_1 * n_2 =	n_1 / n_2 =

Table 2: Question templates for two-operand arithmetic queries.

Formula	Template
$(n_1+n_2)*n_3$	Sum n_1 and n_2 and multiply by n_3
$n_1+n_2*n_3$	What is the sum of n_1 and the product of n_2 and n_3?
$(n_1-n_2)*n_3$	What is the product of n_1 minus n_2 and n_3?
$n_1/(n_2/n_3)$	How much is n_1 divided by the ratio between n_2 and n_3?
$n_1-n_2*n_3$	What is the difference between n_1 and the product of n_2 and n_3?
$n_1*(n_2-n_3)$	How much is n_1 times the difference between n_2 and n_3?

Table 3: Examples of templates of three-operand queries. For the full list, we refer to Karpas et al. (2022).

Model	Operation	Accuracy (%)
GPT-J	+	69.3
	−	78.0
	×	82.8
	÷	40.8
	Overall	67.8
GPT-J (Numeral Words)	+	95.5
	−	86.7
	×	83.3
	÷	59.7
	Overall	81.3
Pythia 2.8B	+	57.4
	−	77.5
	×	64.7
	÷	40.2
	Overall	59.9
LLaMA	+	100.0
	−	99.8
	×	100.0
	÷	88.7
	Overall	97.2
Goat	+	100.0
	−	100.0
	×	91.4
	÷	54.0
	Overall	85.6
Pythia 2.8B (3 Operands)	Overall	0.9
Pythia 2.8B Fine-tuned (3 Operands)	Overall	39.7

Table 4: Accuracy of the models analyzed in the paper on various types of arithmetic queries.

compute

$$\mathbb{1}\{\arg\max_{x\in\mathcal{S}} \mathbb{P}^*_z(x) \neq \arg\max_{x\in\mathcal{S}} \mathbb{P}(x)\}, \quad (4)$$

distinguishing between desired ($\arg\max_{x\in\mathcal{S}} \mathbb{P}_*(x) = r$) and undesired ($\arg\max_{x\in\mathcal{S}} \mathbb{P}(x) = r$) changes. The results reported in Figure 10 show an increase in the desired change in prediction at layers 19-20, while the undesired change in prediction is higher for layers 14-17. This means that interventions on the MLPs at layers 19-20 are more likely to lead to a correct adjustment of the prediction, while the opposite is true for earlier layers (14-15 in particular). This finding is consistent with our previous observations and we see this as additional evidence highlighting the influence of the MLPs at layers 19-20 on the prediction of r.

F Fine-tuning Details

We fine-tune Pythia 2.8B on three-operand queries. We train the model for 2 epochs on a set of queries obtained by sampling 1000 triples of operands for each of the 29 templates. We use Adafactor (Shazeer and Stern, 2018) a learning rate of 10^{-5}, linearly decaying, and a batch size of 8. We make sure that there is no overlap between the set of

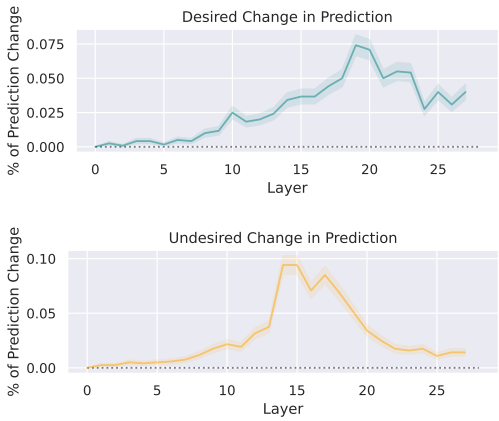

Figure 10: Desired (wrong to correct) and undesired (correct to wrong) change in the prediction induced by the intervention on the MLPs in GPT-J. The layers at which the two types of prediction change peak correspond to the layers with the largest corresponding IE.

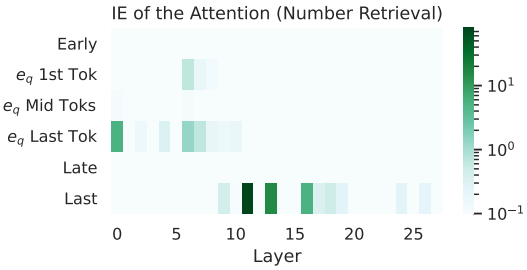

Figure 11: Indirect effect (IE) measured for the attention modules in GPT-J on the task of number retrieval.

queries used for training and the set used for the computation of the indirect effect.

G Computing Infrastructure

The experiments for all models are carried out using a single Nvidia A100 GPU with 80GB of memory. The computation of the indirect effect across the whole model for a single type of component (attention or MLP) took ∼15 hours for GPT-J and ∼6 hours for Pythia (using 50 examples for each two-operand template) and ∼7 hours for LLaMA and Goat (using 20 examples for each two-operand template). For the fine-tuning of Pythia 2.8B, we used a single Nvidia A100 GPU with 80GB of memory. The training procedure took ∼1 hour. Experiment tracking was carried out using Weights & Biases.[8]

[8] http://wandb.ai

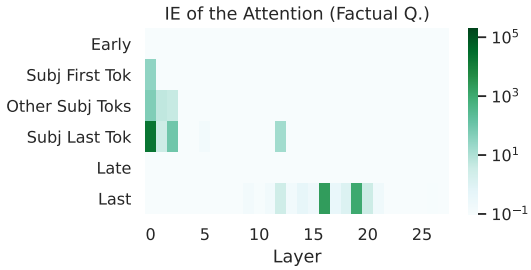

Figure 12: Indirect effect (IE) measured for the attention modules in GPT-J on factual knowledge queries.

H Factual Knowledge Data

For the experiments involving the prediction of factual knowledge, we use the following six relations from the T-REx subset of the LAMA benchmark (Petroni et al., 2019):

1. "[subject] is the capital of [object]"
2. "[subject] was born in [object]"
3. "[subject] died in [object]"
4. "The native language of [subject] is [object]"
5. "[subject] is a subclass of [object]"
6. "The capital of [subject] is [object]".

We sample pairs of subject entities from the set of entities compatible for a given relation (e.g., cities for the relation "is the capital of"). For each relation, we sample 100 pairs of subject entities.

I Log Probability to Quantify the IE

In order to validate whether the measurements of the indirect effect are specific to the metric that we describe in Equation 2, we quantify the IE using the absolute difference in the log of the probability values assigned by the model to the results r and r'. More formally, we compute

$$\mathrm{IE}_{\mathrm{alt}}(z) = \begin{cases} \Delta' + \Delta & \text{if } r \neq r' \\ |\Delta| & \text{otherwise} \end{cases}, \tag{5}$$

where

$$\Delta' = \log \mathbb{P}^*_z(r) - \log \mathbb{P}(r) \tag{6}$$

$$\Delta = \log \mathbb{P}(r') - \log \mathbb{P}^*_z(r'). \tag{7}$$

The results are reported in Figure 13. The activation sites that we observe are the same as reported in Section 4.1: first-layer MLP at the operand tokens and last-token MLP and attention modules.